

Creating tables

Query	Query History
1	CREATE TABLE Airline (
2	airline_id INT PRIMARY KEY ,
3	airline_code VARCHAR(30) ,
4	airline_name VARCHAR(50) NOT NULL ,
5	airline_country VARCHAR(50) NOT NULL ,
6	created_at TIMESTAMP NOT NULL ,
7	updated_at TIMESTAMP NOT NULL
8	);
9	
10	CREATE TABLE Airport (
11	airport_id INT PRIMARY KEY ,
12	airport_name VARCHAR(50) NOT NULL ,
13	country VARCHAR(50) NOT NULL ,
14	state VARCHAR(50) ,
15	city VARCHAR(50) NOT NULL ,
16	created_at TIMESTAMP NOT NULL ,
17	updated_at TIMESTAMP NOT NULL
18	);
19	
20	CREATE TABLE Passengers (
21	passenger_id INT PRIMARY KEY ,
22	first_name VARCHAR(50) NOT NULL ,
23	last_name VARCHAR(50) NOT NULL ,
24	date_of_birth DATE NOT NULL ,

Data Output	Messages	Notifications
CREATE TABLE		
Query returned successfully in 215 msec.		

Total rows:	Query complete 00:00:00.215
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```
Query  Query History
1  INSERT INTO Airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
2  (1, 'A1', 'Air France', 'France', '2023-12-01 10:00:00', NOW()),
3  (2, 'B2', 'Portugal Air', 'Portugal', '2024-01-15 10:00:00', NOW()),
4  (3, 'LOT7', 'Polish Aero', 'Poland', '2024-02-20 10:00:00', NOW()),
5  (4, 'KC9', 'Air Astana', 'Kazakhstan', '2024-04-01 10:00:00', NOW()),
6  (5, 'TK0', 'Turkish Regional Air', 'Turkey', '2024-05-10 10:00:00', NOW()),
7  (6, 'LH', 'Lufthansa', 'Germany', '2024-06-01 10:00:00', NOW());
8
9  INSERT INTO Airport (airport_id, airport_name, country, state, city, created_at, updated_at) VALUES
10 (1, 'Almaty Regional Air Port', 'Kazakhstan', 'Almaty Region', 'Almaty', NOW(), NOW()),
11 (2, 'Astana Reginal Air Hub', 'Kazakhstan', 'Capital', 'Astana', NOW(), NOW()),
12 (3, 'Paris Reginal Air Station', 'France', 'IDF', 'Paris', NOW(), NOW());
13
14 INSERT INTO Passengers (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citi;
15 (1, 'Nurasyl', 'Oralbay', '2004-05-31', 'Male', 'Kazakhstan', 'Kazakhstan', 'N111111', NOW(), NOW()),
16 (2, 'Ali', 'Khan', '1998-10-12', 'Male', 'Kazakhstan', 'Kazakhstan', 'N222222', NOW(), NOW()),
17 (3, 'Diana', 'Saparova', '2006-08-25', 'Female', 'Kazakhstan', 'Kazakhstan', 'N333333', NOW(), NOW());
18
19 INSERT INTO Flights (flight_id, sch_departure_time, sch_arrival_time, departing_airport_id, arriving
20 (101, '2024-05-01 10:00:00', '2024-05-01 14:00:00', 1, 2, 'A1', 'B1', 1, '2024-05-01 10:00:00', '20:
21 (102, '2024-05-02 12:00:00', '2024-05-02 16:00:00', 1, 2, 'A2', 'B2', 2, '2024-05-02 12:00:00', '20:
22 (103, '2024-05-03 15:00:00', '2024-05-03 18:00:00', 2, 3, 'A3', 'B3', 1, '2024-05-03 15:00:00', '20:
23
Data Output  Messages  Notifications
INSERT @ 4
Query returned successfully in 75 msec.
```

Filling in the data in accordance with the conditions of the 4th laboratory.

No limit

Query

Query History

1

--Task 1--

2

SELECT UPPER(airline_name) AS airline_name_uppercase

3

FROM Airline;

Data Output

Messages

Notifications

airline_name_uppercase

text

1

AIR FRANCE

2

PORTUGAL AIR

3

POLISH AERO

4

AIR ASTANA

5

TURKISH REGIONAL AIR

6

LUFTHANSA

Task 1. Retrieve all airline names in uppercase.

Task 2. Replace any occurrence of the word "Air" in airline names with "Aero".

The screenshot shows a SQL IDE interface. The top pane, titled 'Query', contains the following SQL code:

```
1  --Task 2--  
2  SELECT airline_name, REPLACE(airline_name, 'Air', 'Aero') AS modified_airline_name  
3  FROM Airline;
```

The bottom pane, titled 'Data Output', displays the results of the query in a table. The table has two columns: 'airline_name' (character varying (50)) and 'modified_airline_name' (text). The results show six rows of data where the word 'Air' has been replaced by 'Aero' in the airline names.

	airline_name character varying (50)	modified_airline_name text
1	Air France	Aero France
2	Portugal Air	Portugal Aero
3	Polish Aero	Polish Aero
4	Air Astana	Aero Astana
5	Turkish Regional Air	Turkish Regional Aero
6	Lufthansa	Lufthansa

Query Query History

```

1  --Task 3--
2  SELECT flight_id, airline_id
3  FROM Flights
4  WHERE airline_id IN (1, 2);

```

Data Output Messages Notifications

SQL query of data

	flight_id [PK] integer	airline_id integer
1	101	1
2	102	2
3	103	1

Task 3. Find all flight numbers that coordinates with both airline 1 and airline 2.

Query Query History

```

1  --Task 4--
2  SELECT *
3  FROM Airport
4  WHERE airport_name LIKE '%Reginal%'
5  AND airport_name LIKE '%Air%';

```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	airport_id [PK] integer	airport_name character varying (50)	country character varying (50)	state character varying (50)	city character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	2	Astana Reginal Air Hub	Kazakhstan	Capital	Astana	2026-10-05 09:38:58.134599	2026-10-05 09:38:58.134599
2	3	Paris Reginal Air Stati...	France	IDF	Paris	2026-10-05 09:38:58.134599	2026-10-05 09:38:58.134599

Task 4. Retrieve airports that contain the word "Reginal" and "Air" in their names.

Task 5. Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

QueryQuery History

```
1  --Task 5--
2  SELECT
3      first_name || ' ' || last_name AS passenger_name,
4      TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS formatted_dob
5  FROM Passengers;
```

Data OutputMessagesNotifications



Showing 1

	passenger_name text	formatted_dob text
1	NurasyI Oralbay	May 31, 2004
2	Ali Khan	October 12, 19...
3	Diana Saparova	August 25, 2006

Query

Query History

1

--Task 6-7--

2

SELECT flight_id, sch_arrival_time, act_arrival_time,

3

(act_arrival_time - sch_arrival_time) AS delay_duration

4

FROM Flights

5

WHERE act_arrival_time > sch_arrival_time;

Data Output

Messages

Notifications

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SQL

Showing rows:

	flight_id [PK] integer	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone	delay_duration interval
1	101	2024-05-01 14:00:00	2024-05-01 15:30:00	01:30:00
2	103	2024-05-03 18:00:00	2024-05-03 20:00:00	02:00:00

Task 6-7. Find flight numbers that have been delayed based on the actual arrival time& Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

1

--Task 8--

2

SELECT *

3

FROM Airline

4

WHERE airline_country IN ('France', 'Portugal', 'Poland')

5

AND created_at BETWEEN '2023-11-01' AND '2024-03-31';

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 3

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Page No: 1 of 1

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	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	A1	Air France	France	2023-12-01 10:00:00	2026-10-05 09:38:58.134599
2	2	B2	Portugal Air	Portugal	2024-01-15 10:00:00	2026-10-05 09:38:58.134599
3	3	LOT7	Polish Aero	Poland	2024-02-20 10:00:00	2026-10-05 09:38:58.134599

Task 8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

Find top3 overweighted baggage with more than 25kg.

Query

Query History

1

--Task 9--

2

SELECT *

3

FROM Baggage

4

WHERE weight_in_kg > 25

5

ORDER BY weight_in_kg DESC

6

LIMIT 3;

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 3

baggage_id

[PK] integer

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weight_in_kg

numeric (5,2)

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created_at

timestamp without time zone

🖋️

updated_at

timestamp without time zone

🖋️

booking_id

integer

🖋️

1

2

32.00

2026-10-05 09:38:58.134599

2026-10-05 09:38:58.134599

2

2

5

30.00

2026-10-05 09:38:58.134599

2026-10-05 09:38:58.134599

300

3

1

28.50

2026-10-05 09:38:58.134599

2026-10-05 09:38:58.134599

1

Task 10. Find the youngest passengers' full name.

Query

Query History

1

--Task 10--

2

SELECT first_name || ' ' || last_name AS youngest_passenger, date_of_birth

3

FROM Passengers

4

ORDER BY date_of_birth DESC

5

LIMIT 1;

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 1

	youngest_passenger text	date_of_birth date
1	Diana Saporova	2006-08-25

Task 11. Find the cheapest booking price on each booking platform.

QueryQuery History

1--Task 11--

2SELECT booking_platform, MIN(ticket_price) AS cheapest_price

3FROM Booking

4GROUP BY booking_platform;

Data OutputMessagesNotifications

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SQL

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	booking_platform character varying (50) 🔒	cheapest_price numeric 🔒
1	Website	9500.00
2	Agent	8000.00
3	Mobile App	11000.00

Task 12. Return airlines whose airline_code contains a digit.

QueryQuery History

1

--Task 12--

2

SELECT *

3

FROM Airline

4

WHERE airline_code ~ '[0-9]';

Data OutputMessagesNotifications						
Showing rows: 1 to 5Page No: 1 of 1						
	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	A1	Air France	France	2023-12-01 10:00:00	2026-10-05 09:38:58.134599
2	2	B2	Portugal Air	Portugal	2024-01-15 10:00:00	2026-10-05 09:38:58.134599
3	3	LOT7	Polish Aero	Poland	2024-02-20 10:00:00	2026-10-05 09:38:58.134599
4	4	KC9	Air Astana	Kazakhstan	2024-04-01 10:00:00	2026-10-05 09:38:58.134599
5	5	TK0	Turkish Regional Air	Turkey	2024-05-10 10:00:00	2026-10-05 09:38:58.134599

Task 13. List the top5 most recently created airlines.

QueryQuery History

1

--Task 13--

2

SELECT *

3

FROM Airline

4

ORDER BY created_at DESC

5

LIMIT 5;

Data OutputMessagesNotifications						
Showing rows: 1 to 5Page No: 1 of 1						
	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	6	LH	Lufthansa	Germany	2024-06-01 10:00:00	2026-10-05 09:38:58.134599
2	5	TK0	Turkish Regional Air	Turkey	2024-05-10 10:00:00	2026-10-05 09:38:58.134599
3	4	KC9	Air Astana	Kazakhstan	2024-04-01 10:00:00	2026-10-05 09:38:58.134599
4	3	LOT7	Polish Aero	Poland	2024-02-20 10:00:00	2026-10-05 09:38:58.134599
5	2	B2	Portugal Air	Portugal	2024-01-15 10:00:00	2026-10-05 09:38:58.134599

Query

Query History

1

--Task 14--

2

SELECT *

3

FROM Baggage_check

4

WHERE booking_id BETWEEN 200 AND 300

5

AND check_result <> 'Checked';

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 2

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Page No: 1

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	1	Not checked	2024-05-01 10:00:00	2026-10-05 09:38:58.134599	250	3
2	2	Pending	2024-05-01 11:00:00	2026-10-05 09:38:58.134599	251	1

Task 14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

Query

Query History

1

--Task 15--

2

SELECT *

3

FROM Baggage_check

4

WHERE EXTRACT(MONTH FROM updated_at) = EXTRACT(MONTH FROM created_at)

5

AND EXTRACT(YEAR FROM updated_at) = EXTRACT(YEAR FROM created_at)

6

AND updated_at < created_at;

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1

	baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
1	4	Pending	2024-03-20 15:00:00	2024-03-20 10:00:00	1	1

Task 15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.